# Ahmed Ikram | Data Engineer

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Final-year BSc Computer Science (Data Science & AI) student at the University of Dundee (predicted First, graduating June 2027). Built batch lakehouse and streaming platforms for BI and ML consumers, with rigorous quality gates, lineage, and cost control. Seeking a graduate Data Engineering role from June 2027.

## EDUCATION

**University of Dundee** - *BSc (Hons) Computer Science (Data Science and AI)* *Sep 2023 - Present*

- On track for a First-Class Honours degree. Expected Graduation: June 2027

**High School of Dundee** *Sep 2018 - June 2023*

- Highers: A Mathematics, A Computing Science, A English, B Chemistry
- Computing Science classroom assistant - mentored junior pupils one-to-one during classwork.

## TECHNICAL SKILLS

**Languages & Stores:** Python (pandas, NumPy, PyArrow) · SQL · T-SQL · PostgreSQL · Azure SQL (SQL Server) · Redis · Alembic

**Orchestration & Transformation:** Apache Airflow · Apache Kafka · dbt · Apache Spark / PySpark · Pydantic · MLflow

**Streaming & Warehousing:** OpenLineage · Marquez · ClickHouse · Delta Lake · BigQuery · Databricks (DLT, Unity Catalog) · Azure SQL · Power BI

**Data Quality & Observability:** Great Expectations · Grafana · Prometheus · Azure Monitor · Cloud Monitoring · CloudWatch · StatsD · Evidently AI

**Cloud & DevOps:** AWS (ECS Fargate, EC2, S3, RDS, ElastiCache, WAF, IAM, CloudFront, Secrets Manager) · Azure (Databricks, Azure SQL, ADLS) · Google Cloud (GCS, BigQuery) · Terraform · Docker · GitHub Actions · Linux

## PROJECTS

**W3C Web Logs ETL Pipeline - ADLS to Power BI**

*Airflow · Databricks · dbt · PySpark · Azure SQL · Delta Lake · Terraform · Power BI · Marquez · Grafana · Prometheus*

- Databricks DLT medallion architecture (Bronze/Silver/Gold) pipeline with 7 DLT quality gates ingests (153,380 rows, 0 dropped); a consolidated MaxMind GeoIP struct UDF runs 3.5× faster than 7 separate UDFs and reaches 99.99% known-country coverage across 153,377 enriched rows.
- 16 dbt models as a Kimball star schema (SCD Type 1 dimensions; SCD Type-2 on GeoLocation) on hashed natural keys, compiling to PostgreSQL and T-SQL from one codebase via 18 compatibility macros; 121 dbt tests gate every model before Power BI and catch 213 duplicate rows on a 12-component incremental deduplication key.
- 2 Airflow DAGs orchestrate Databricks with Dataset-triggered decoupling of ingest from transform; OpenLineage → Marquez records the cross-engine graph (DLT → Azure SQL → dbt → Power BI) that Unity Catalog cannot see, including dbt test outcomes on the producing node; a 2-part Terraform estate (30+ Azure resources, OIDC federation, rollback workflow) deploys the platform.
- JDBC export of 153,377 Silver rows to Azure SQL dropped from 413s to 45s under serverless DLT's no-JDBC-write constraint; a freshness probe and 3 Grafana dashboards (23 panels, 10 alert rules) back a 7-page Power BI report (156K requests, 88 countries), with 627 tests across 5 tiers including dual-dialect dbt CI.

**WikiStream Real-Time Streaming Analytics**

*Python · ClickHouse · Grafana · BigQuery · Grafana · Great Expectations · Terraform · Google Cloud · Pydantic*

- Async consumer of Wikipedia's EventStreams feed into a self-hosted ClickHouse cluster - 58.9M+ events in 3 days, real peaks of 2,719/sec; Pydantic schema-on-write rejects malformed events to a dead-letter table before they hit disk.
- At-least-once delivery with a durable cursor and a deduplication ring buffer, proven with a live SIGKILL-mid-insert test (zero loss, zero duplicates); an 8-injection chaos battery fired and cleared every Grafana and Cloud Monitoring rule.
- Three materialized views cut dashboard query latency 15x at p50 (13.2x at p99) and rows scanned ~200x, holding 5,655 events/sec (2.08x real peak) with zero drops proving scaling headroom; hourly ClickHouse → GCS → BigQuery export keeps a 30-day-plus warehouse tier, parity-checked by exact-sum comparison every run.

- Great Expectations gates freshness, nulls, cardinality and schema drift hourly; GCP via Terraform + WIF, itemised FinOps tracking held to $41.65/month run-rate, validated by 143 tests at 99.4% coverage.

**ATM Log Aggregation, Anomaly Detection & Diagnostics - Event Pipeline for NCR Atleos**

*Apache Kafka · FastAPI · PostgreSQL · Redis · MLflow · Docker · Terraform · AWS · SageMaker*

- Replaced a direct-DB generator with a distributed Kafka (KRaft) event pipeline: at-least-once delivery with idempotent deduplication, hybrid Redis SET+TTL and LRU fallback, Redis Stream DLQ with exponential backoff, sustaining 100 msg/s over 2.5M events/day across 7 log sources.
- PostgreSQL data lake on JSONB payloads - new sources only need a new parser, zero migrations; 14 indexes (6 composite) hold sub-100ms queries past 2M+ rows under a pooled, retrying connection layer, and feed a downstream detector with a SageMaker cross-check.
- MLflow tracks every training run with champion-alias promotion, so ML consumers read a gated artifact; AWS estate of 10 Terraform modules / 118 resources (multi-AZ VPC, ECS Fargate, RDS, S3, CloudFront, 6 least-privilege IAM roles) with ingestion tests catching data loss under 50 concurrent writes.

**StockLens Feature & Retraining Pipeline**

*Airflow · AWS · PyTorch · MLflow · Evidently AI · PostgreSQL · Redis · Terraform*

- Weekly Airflow DAG; OHLCV ingest → 17-feature build → MLflow-tracking training → champion/challenger promotion and blocks any ship that does not beat the live champion by >2pp so a regression cannot reach consumers.
- Evidently AI drift monitors PSI/KS/JSD on live vs training distributions beside the DAG and alerts to CloudWatch before drift reaches the champion - the same quality-gate pattern as the data pipelines, applied to a model artifact.
- Champion delivery writes the gated `model.pt` to EFS (zero-copy inference) with S3 and a PostgreSQL model registry, so serving never reads an unpromoted run; prediction logs retain 90 days for offline replay.

## CERTIFICATES

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|---|---|
| AWS Academy Graduate - Machine Learning Foundations | *Feb 2025* |
| Foundations of Azure AI: Concepts, Capabilities, and Implementation | *Oct 2024* |
| AWS Academy Graduate - Cloud Foundations | *Mar 2024* |

## EXPERIENCE

**Beauly Mini Market,** Dundee, Scotland

*Retail Assistant* — *September 2023 - Present*

**Zulus Peri Peri Grill,** Dundee, Scotland

*Front of House Worker* — *June 2021 - September 2023*